ASSIGNMENT 1 – CO1 | TASK 12

Design Challenge: Airport, Weekend Discount and Monthly Pass

Scenario: Namma Metro Smart-Card Fare System

Objective / explanation

The modified design adds Airport (Rs 60), applies a 10% weekend discount instead of the peak surcharge, and models a monthly pass valid for two selected zones.

C++ source code

```
#include &lt;array&gt;
#include &lt;iostream&gt;
#include &lt;set&gt;
using namespace std;
enum class Zone { Inner=0, Middle=1, Outer=2, Airport=3 };
constexpr array&lt;long long,4&gt; BASE{1000,2000,3000,6000};

long long calculateFare(Zone z, int hour, bool weekend) {
    int i = static_cast&lt;int&gt;(z);
    if (i &lt; 0 || i &gt;= static_cast&lt;int&gt;(BASE.size()) || hour &lt; 0 || hour &gt; 23)
        return -1;
    long long base = BASE[static_cast&lt;size_t&gt;(i)];
    if (weekend) return base - base * 10 / 100;
    bool peak = (hour &gt;= 8 &amp;&amp; hour &lt; 10) || (hour &gt;= 17 &amp;&amp; hour &lt; 20);
    return peak ? base + base * 20 / 100 : base;
}
class MonthlyPass {
    set&lt;Zone&gt; zones;
public:
    MonthlyPass(Zone a, Zone b) : zones{a,b} {}
    bool covers(Zone z) const { return zones.count(z) != 0; }
};
int main() {
    cout &lt;&lt; "Airport off-peak paise: " &lt;&lt; calculateFare(Zone::Airport,12,false) &lt;&lt; '\n';
    cout &lt;&lt; "Airport peak paise: " &lt;&lt; calculateFare(Zone::Airport,18,false) &lt;&lt; '\n';
    cout &lt;&lt; "Airport weekend paise: " &lt;&lt; calculateFare(Zone::Airport,18,true) &lt;&lt; '\n';
    MonthlyPass pass(Zone::Inner, Zone::Middle);
    cout &lt;&lt; "Pass covers Inner: " &lt;&lt; pass.covers(Zone::Inner) &lt;&lt; '\n';
    cout &lt;&lt; "Pass covers Airport: " &lt;&lt; pass.covers(Zone::Airport) &lt;&lt; '\n';
}
```

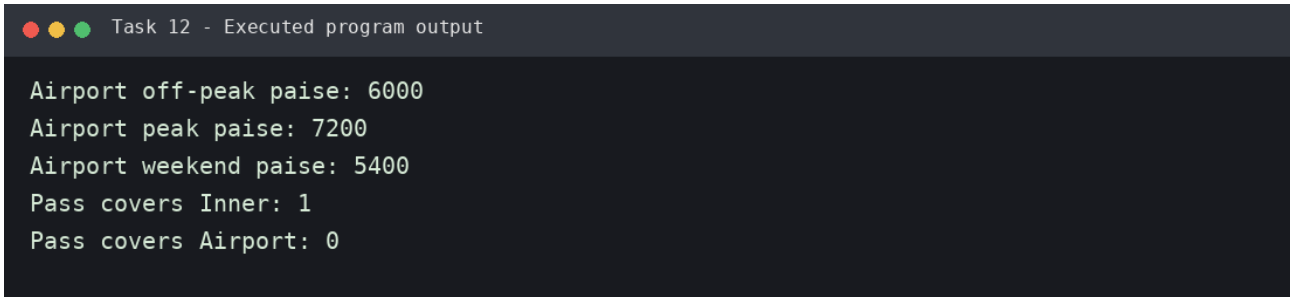
Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

Actual captured program output

```
Airport off-peak paise: 6000
Airport peak paise: 7200
Airport weekend paise: 5400
Pass covers Inner: 1
Pass covers Airport: 0
```

Output screenshot



Analysis / complexity

Decision: modify the design. A four-entry fare table is easier to extend, weekend and peak rules are separate, and MonthlyPass handles a distinct entitlement. Production code should also track pass expiry and validate route/zone policy.

Evidence note: Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.